

The $d = 2$ cutoff expansion with an explicit remainder constant

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Abstract

Astra #9 R5, step 2. Every step of the $d = 2$ cutoff assembly is now quantitative for $N \geq 1$: the face transfers carry the explicit constant `faceConst` (an envelope moment plus tail moments of the face coefficients, read off `face_transfer_bound`), the mixed remainder carries the Gaussian-moment constant of unit 130, and a bound at a larger exponent is a bound at the target exponent for $N \geq 1$. Summing,

$$|Z(N) - S^{\text{merged}}(N)| \leq \text{cutoffConst} \cdot N^{-E} (1 + \log N), \quad E = \min\left(p_1 + \frac{M_1}{k_1}, p_2 + \frac{M_2}{k_2}\right),$$

for every $N \geq 1$ and arbitrary starting exponents (`twoD_cutoff_general_explicit`), with `cutoffConst` the finite sum of the face, corner and mixed constants. Zero `sorry/axiom`.

1 The things formalised

```
theorem bound_mono_exp (X K q E N : ) (hN : 1 ≤ N) (hqE : E ≤ q)
  (h : |X| ≤ K * N ^ (-q) * (1 + Real.log N)) : |X| ≤ K * N ^ (-E) * (1 + Real.log N)
noncomputable def faceConst ( p b : ) (h k k M : ) (Ψ : → → ) (f : → → )
  (H : → ) : :=
  envMoment (p + ((M : ) - ((k : ) * p - h - 1)) / k) 1
  (faceEnv ((k : ) * p - h - 1) (b ^ k) k k M H)
  + i, (b ^ (k + k)) ^ (face p ((k : ) * p - h - 1) k M i
    - (p + ((M : ) - ((k : ) * p - h - 1)) / k))
  * tailMoment (p + ((M : ) - ((k : ) * p - h - 1)) / k) (faceJ M i)
  (faceC ((k : ) * p - h - 1) b k k Ψ f M i)
theorem face_bound_explicit ... (N : ) (hN : 1 ≤ N) :
  |twoDamp b N h h k k (fun u _ s => Ψ u s) - faceSum p b h k k M Ψ f N|
  faceConst p b h k k M Ψ f H
  * N ^ (-(p + ((M : ) - ((k : ) * p - h - 1)) / k)) * (1 + Real.log N)
noncomputable def cutoffConst ( b L C : ) (h h k k M M D : ) (a bj) (c) (Ha Hb) :
  :=
  ( i range M, faceConst (u-face i)) + ( j range M, faceConst (v-face j))
  + ( i range M, j range M, faceConst (corner i j))
  + C * Real.exp ((D : ) ^ 2 / ) * mixConst ( / 2) (2 * L) b (h + M) (h + M) k k
theorem twoD_cutoff_general_explicit ... (same hypotheses as twoD_cutoff_general_isBig0)
  (N : ) (hN : 1 ≤ N) :
  |twoDamp b N h h k k Φ - mergedSum b h h k k M M a bj c N|
  cutoffConst b L C h h k k M M D a bj c Ha Hb
  * N ^ (-(min (p + (M : ) / k) (p + (M : ) / k))) * (1 + Real.log N)
```

2 Mathematics

The decomposition $Z - S^{\text{merged}} = \sum_i F_i^A + \sum_j F_j^B - \sum_{ij} F_{ij}^C + F^R$ is the rectangular identity of unit 106 combined with the over-complete-to-merged identity of unit 127 (at the standard corner order). Each face piece is bounded by `face_transfer_bound` at its own exponent ($p_2 + M_2/k_2$ for u -faces, $p_1 + M_1/k_1$ for v -faces, $p_1 + (M_1 + k_1 M_2 + i)/k_1$ for corners), the mixed piece by unit 130 at $\min(p_1 + M_1/k_1, p_2 + M_2/k_2)$; all exponents are $\geq E$, and for $N \geq 1$ a bound $|X| \leq K N^{-q}(1 + \log N)$ with $q \geq E$ implies $|X| \leq K N^{-E}(1 + \log N)$ (the constant is nonnegative because the right side dominates $|X| \geq 0$ with a positive factor). The triangle inequality over the finite sums gives the sum of constants.

3 Paper-facing meaning

With unit 130 this is the “explicit cutoff bound” half of Astra #9’s R5: the remainder in `thm:TaylorTree` ($d = 2$) is bounded by a named finite expression for every $N \geq 1$, not only eventually. For the uniform-family theorem the face constants must still be bounded in terms of the envelope data alone: `faceConst` involves the tail moments of the actual face coefficients ($k_1^{-1}\text{FP}$, monomial coefficients, log coefficient), which the envelope constructor of unit 110 bounds by $(1 + s)^D e^{\beta s a'}$ envelopes; turning those pointwise bounds into moment bounds is the next step.

4 Lean notes

- `face_transfer_bound` is already stated for the swapped amplitude, so only the goal needs the `twoDamp_pow_absorb/twoDamp_swap` rewrite, not the hypothesis.
- `set A := N ^ (-E) * (1 + log N)` does not fold later-appearing copies; rewrite with `hAdef` and `← mul_assoc` to line up `K * A` with the bounds’ shape `K * N ^ (-E) * (1 + log N)`.
- Nonnegativity of an explicit constant is cheapest to extract from the bound it appears in: `le_of_mul_le_mul_right` with `0 * A` `K * A`.
- `Finset.sum_mul` then `Finset.abs_sum_le_sum_abs` and `Finset.sum_le_sum` turn termwise bounds `|F i|` `K i * A` into a bound on the absolute value of the sum.